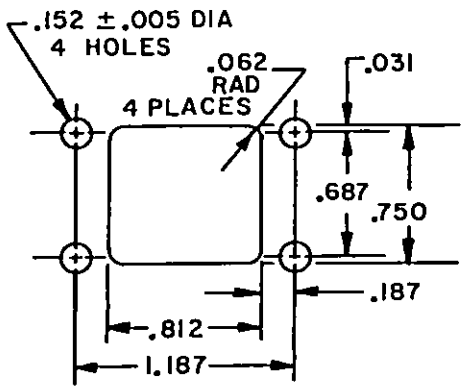
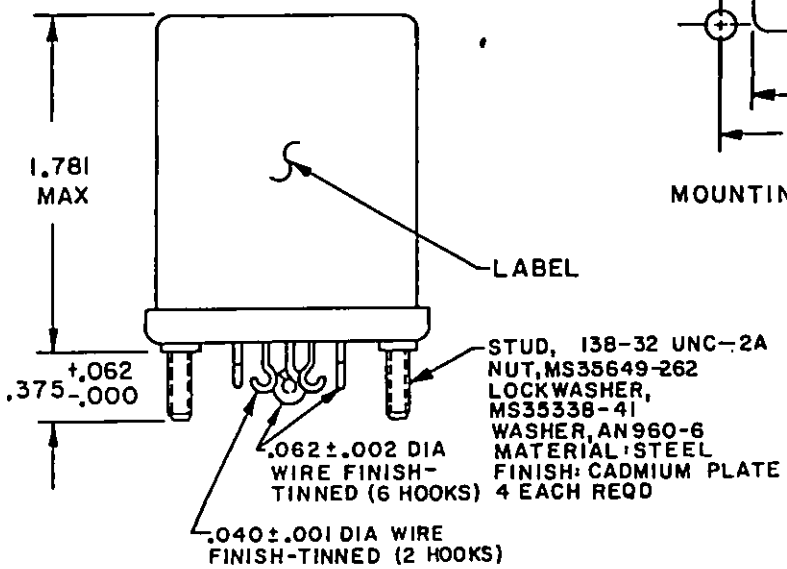
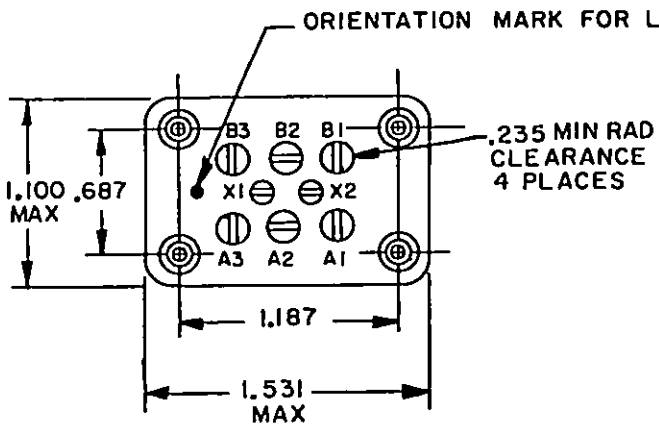
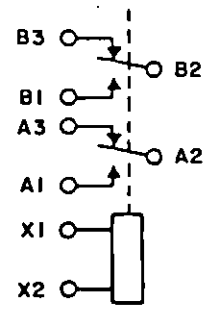
FED. SUP CLASS
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MOUNTING HOLE LAYOUT







CIRCUIT DIAGRAM
(SEE NOTE 7)

Ⓖ ENTIRE STANDARD REVISED

APPROVED 23 June 1959 REVISED Ⓖ 5 JUN 87

User activities: Army - Navy - Air Force -

Review activities: Army - Navy - EC Air Force - 11, 99

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

P.A USAF - 85 Other Cust Navy - AS	International Interest	TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2 PDT, TYPE I, SOLDER TERMINALS, STUD MOUNTED, HERMETICALLY SEALED	MILITARY STANDARD MS25273
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 1 OF 5

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Inches	mm
.001	0.03
.002	0.05
.005	0.13
.010	0.25
.031	0.79
.040	1.02
.062	1.57
.138	3.51
.152	3.85
.187	4.75
.235	5.97
.357	9.53
.687	17.45
.750	19.05
.812	20.62
1.100	27.94
1.187	30.15
1.531	38.89
1.781	45.24

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Terminal numbers need not appear on relay headers provided there is affixed to the relay a suitable legible circuit diagram that permanently and positively identifies each terminal location specified herein.
5. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
6. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.
7. The use of diodes on ac relays is optional. Actual application must be shown on label (dash numbers -A1 and -A2 are inactive for new design).

TABLE I. Dash numbers and characteristics.

Dash number MS25273-	Type	Coil	Terminal type	Mounting	Max weight in pounds
D1	I	dc	Solder hook	Stud	0.30
A1 1/	I	ac	Solder hook	Stud	0.32
A2 1/	I	ac	Solder hook	Stud	0.32

1/ Dash number -A1 and -A2 are inactive for new design and shall be used for support of existing systems only.

APPROVED 23 June 1959 REVISED (G) ENTIRE STANDARD REVISED.

User activities: Army - Navy - Air Force -
Review activities: Army - EC Navy - Air Force - 11, 99
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Review activities: Army -
Navy - EC
Air Force - 11, 99

User activities: Army -
Navy -
Air Force -

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Procurement Specification MIL-R-6106

P.A USAF - 85
Other Cust Navy - AS

International interest

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2 PDT, TYPE I, SOLDER TERMINALS STUD MOUNTED, HERMETICALLY SEALED

MILITARY STANDARD

MS25273

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TABLE II. Operating characteristics.

MS part no. MS25273-	Coil data											Time - (milliseconds maximum)					
	Coil	Nominal			Max		Max pick-up voltage			Drop-out voltage 2/	Hold voltage 2/	Operate 3/	Release 4/	Contact bounce			
		Volts 1/	Freq. Hz	Res Ω ±10%	Volts	Amperes	Normal 2/	High temp test	Cont current test					Main		Aux	
														NO	NC	NO	NC
D1	X1, X2	28	dc	175	29	0.20	18	19.5	22.5	1.5	7.0	20	20	2	2		
A1 5/ 6/	X1, X2	115	400	N/A	122	0.07	90	95	103	5.0	30	25	50	2	2		
A2 6/	X1, X2	115	50/60	N/A	122	0.10	90	95	103	5.0	30	25	50	2	2		

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ MS25273-A1 may be used on 60 Hz if maximum ambient temperature is limited to 85°C (maximum coil current shall be 0.077 ampere).
6/ Inactive for new design.

APPROVED 23 June 1958 REVISED (G) ENTIRE STANDARD REVISED.

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Review activities: Army -
Navy - EC
Air Force - 11, 99

User activities: Army -
Navy -
Air Force -

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P.A
USAF - 85
Navy - AS
Other Cust

Procurement Specification
MIL-R-6106

International Interest

SUPERSEDES:

TITLE
RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2 PDT, TYPE I, SOLDER TERMINALS STUD MOUNTED, HERMETICALLY SEALED

MILITARY STANDARD

MS 25273

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FED. SUP CLASS
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TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles X 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	10	10			10	6			10	6			
Inductive	100													
Inductive	20	6	6			6	4			6	4			
Motor	100	4	4			4	3			4	3			
Lamp	100	2	2			2	1.5			2	1.5			
Transfer load														2/
Mechanical life reduced current	400	2.5	2.5			2.5	2			2.5	2			
Intmd current	Applicable per specification													

1/ Absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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Environmental characteristics

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	50 G
Duration	11 ms
Max duration contact opening	10 µs
Vibration - sinusoidal	
G-level	10 G
Frequency range	5 - 1500 Hz
Acceleration	15 G

Electrical characteristics

Insulation resistance	100 megohms
After life or environmental tests	50 megohms

Dielectric strength (sea level)

	Initial	After life tests
Coil to case	1,000 V rms	1,000 V rms
Aux contacts	1,500 V rms	1,125 V rms
All other points		

Dielectric strength (altitude)

	80,000 ft
Coil to case	250 V rms
Aux contacts	
All other points	350 V rms

Max contact drop initial	0.150 volt
After life test	0.175 volt
Overload current	40 amperes dc, 60 amperes ac
Rupture current	50 amperes dc, 80 amperes ac
Duty rating	Continuous
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

Quality conformance inspection

Performance of groups B and C tests not applicable to dash numbers -A1 and -A2.

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